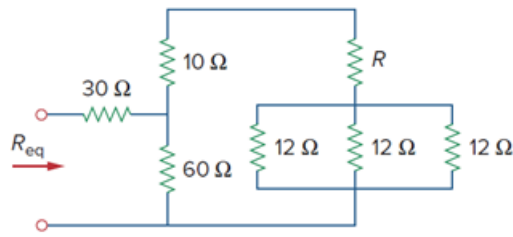


Problem

If $R_{eq} = 50\ \Omega$ in the circuit of Fig. 2.105, find R .

Figure 2.105:



Step-by-step solution

Step 1 of 5

Draw the circuit diagram.

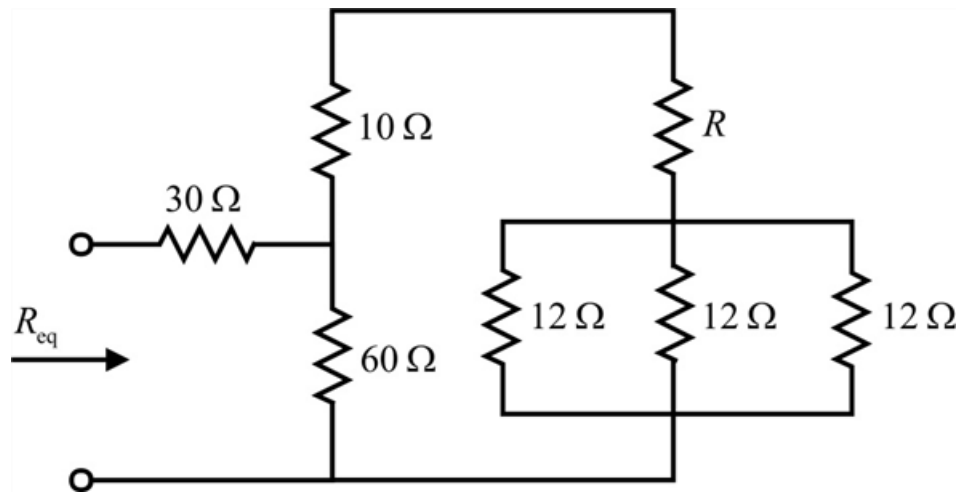


Figure 1

Step 2 of 5

In Figure 1, the three $12\ \Omega$ resistors are connected in parallel. First, determine its equivalent resistance.

$$\begin{aligned}\frac{1}{R_1} &= \frac{1}{12} + \frac{1}{12} + \frac{1}{12} \\ &= \frac{1+1+1}{12} \\ &= \frac{3}{12} \\ \frac{1}{R_1} &= \frac{1}{4} \\ R_1 &= 4\ \Omega\end{aligned}$$

Step 3 of 5

Now, draw the modified circuit diagram.

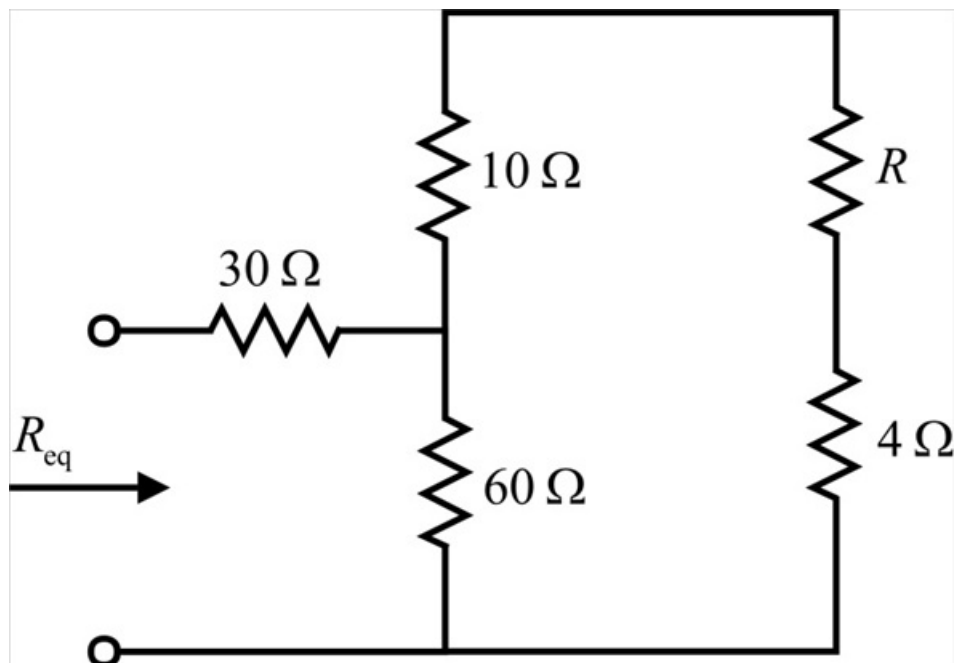


Figure 2

Step 4 of 5

In Figure 2, the resistors $10\ \Omega$, R and $4\ \Omega$ are connected in series. Its equivalent resistance becomes $14 + R\ \Omega$. Now, draw the reduced circuit diagram.

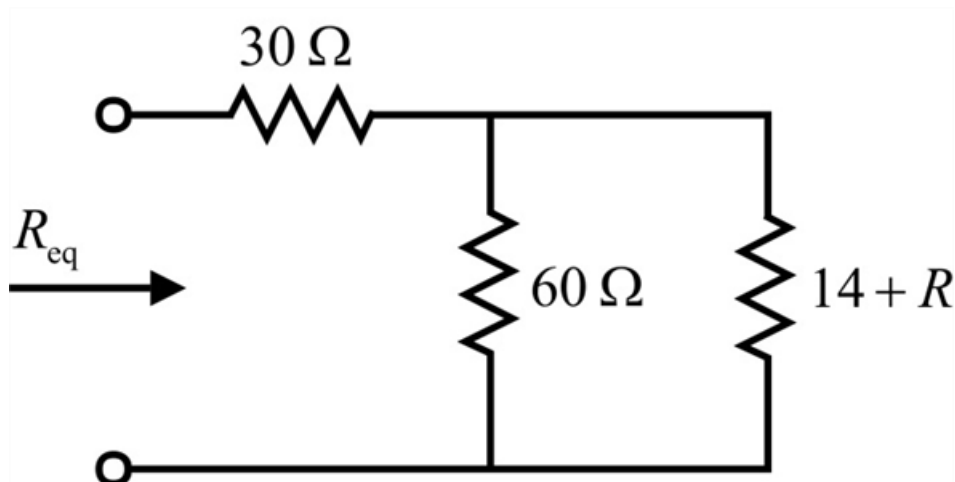


Figure 3

Step 5 of 5

In Figure 3, the parallel combination of resistors $60\ \Omega$ and $14 + R\ \Omega$ is in series with the $30\ \Omega$ resistor. Now, determine the equivalent resistance of the circuit.

$$R_{eq} = 30 + (60) \parallel (14 + R)$$

$$R_{eq} = 30 + \frac{(60)(14 + R)}{60 + 14 + R} \dots\dots (1)$$

Substitute $50\ \Omega$ for R_{eq} in equation (1).

$$50 = 30 + \frac{(60)(14 + R)}{60 + 14 + R}$$

$$\frac{(60)(14 + R)}{74 + R} = 20$$

$$\frac{14 + R}{74 + R} = \frac{20}{60}$$

$$\frac{14 + R}{74 + R} = \frac{1}{3}$$

$$42 + 3R = 74 + R$$

$$2R = 32$$

$$R = 16\ \Omega$$

Therefore, the value of resistor, R is $16\ \Omega$.